

Overview & Impact

Research Case

Datasets

Indices

Interpretation

Conclusion



LANDSLIDES

AND THEIR GOVERNING FACTORS

Research

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Landslide

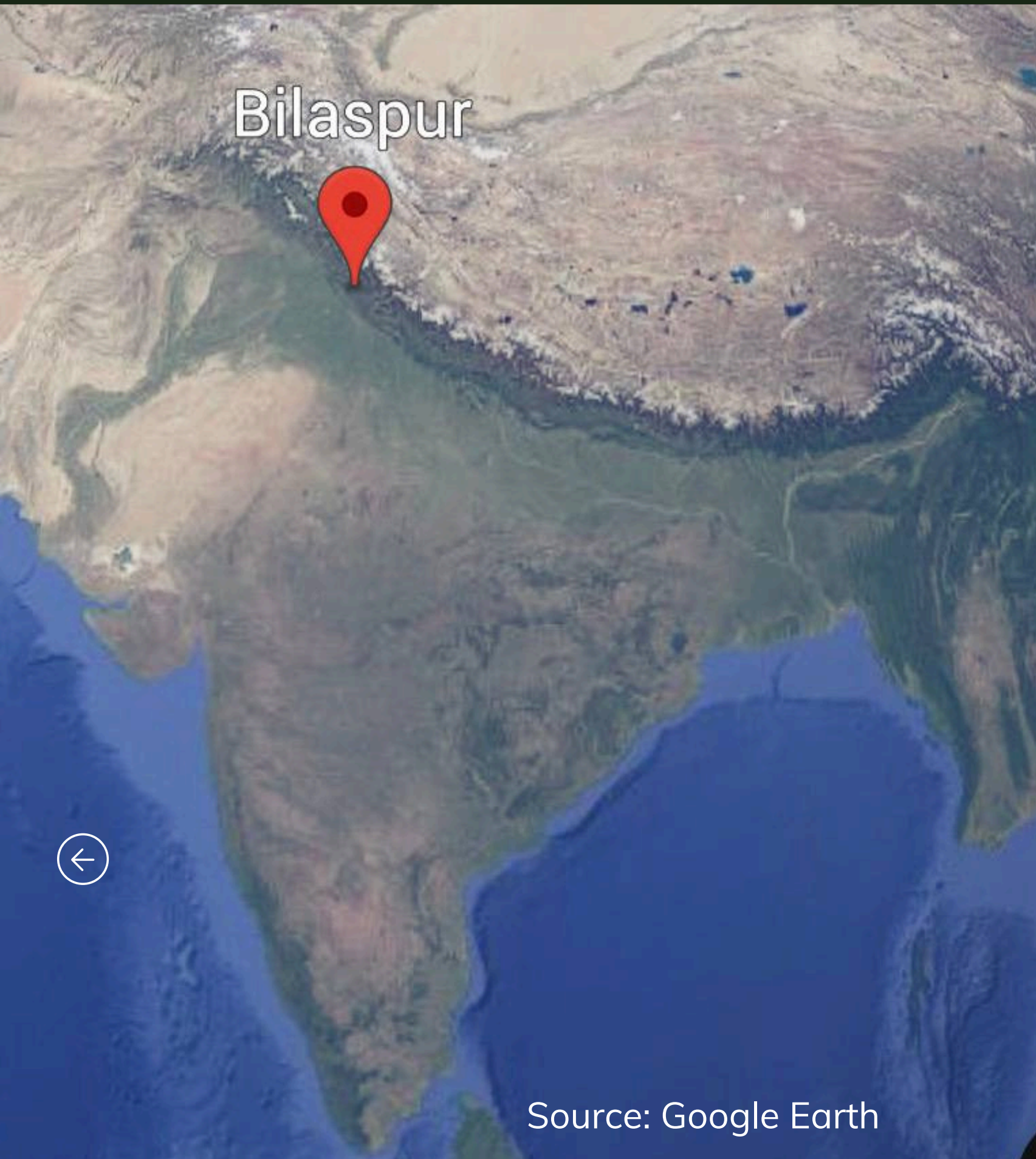
A landslide is the downward movement of soil, rock, or debris along a slope due to gravity. It occurs when slope stability is reduced by factors such as heavy rainfall, earthquakes, volcanic activity, or human actions like deforestation and construction. Landslides may happen suddenly or develop slowly and are common in hilly and mountainous regions. They pose serious threats to human life, infrastructure, and the natural environment.

World Impact

Landslides cause major global damage, leading to loss of life, destruction of homes, roads, and farmland, and serious economic losses. They displace communities, disrupt transport and communication, and can block rivers, causing floods and environmental damage. Climate change-driven increases in intense rainfall have made landslides more frequent worldwide.

Landslides in India

- Landslides are a severe natural hazard in India, mainly affecting the Himalayas, Western Ghats, and Northeast India, especially during the monsoon season.
- The Bilaspur landslide (recent) showed high severity, where heavy rainfall triggered slope failure, causing loss of lives, road blockage, and disruption of transport.
- The Kedarnath landslide (2013) caused massive destruction of settlements and infrastructure due to extreme rainfall.
- The Malin landslide (2014) buried an entire village, highlighting the deadly impact on human life.
- Overall, landslides in India cause loss of life, damage to property, economic losses, and long-term environmental degradation.



Source: Google Earth

Case Research

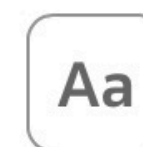
Bilaspur district
Himachal Pradesh, India.

Pre Event Dataset:
Post Event Dataset:

Death toll climbs to 15 after landslide hits bus in India's Himachal Pradesh

By Reuters

October 8, 2025 10:00 AM GMT+5:30 · Updated October 8, 2025



NEW DELHI, Oct 8 (Reuters) - The death toll from a landslide that struck a private bus in India's mountainous state of Himachal Pradesh has risen to 15, with one child still missing, officials and police said on Wednesday.

Source: Reuters | Screenshot of news report

02

Date: October 7th 2025

Area: Bhalughat

Sub-division: Jhanduta

Fatalities: 15

A large volume of rocks, soil, and debris suddenly moved downslope onto the roadway. The landslide caused severe damage to the road infrastructure and completely blocked traffic in the area. The event highlighted the high vulnerability of Himalayan regions to slope failure and land instability.

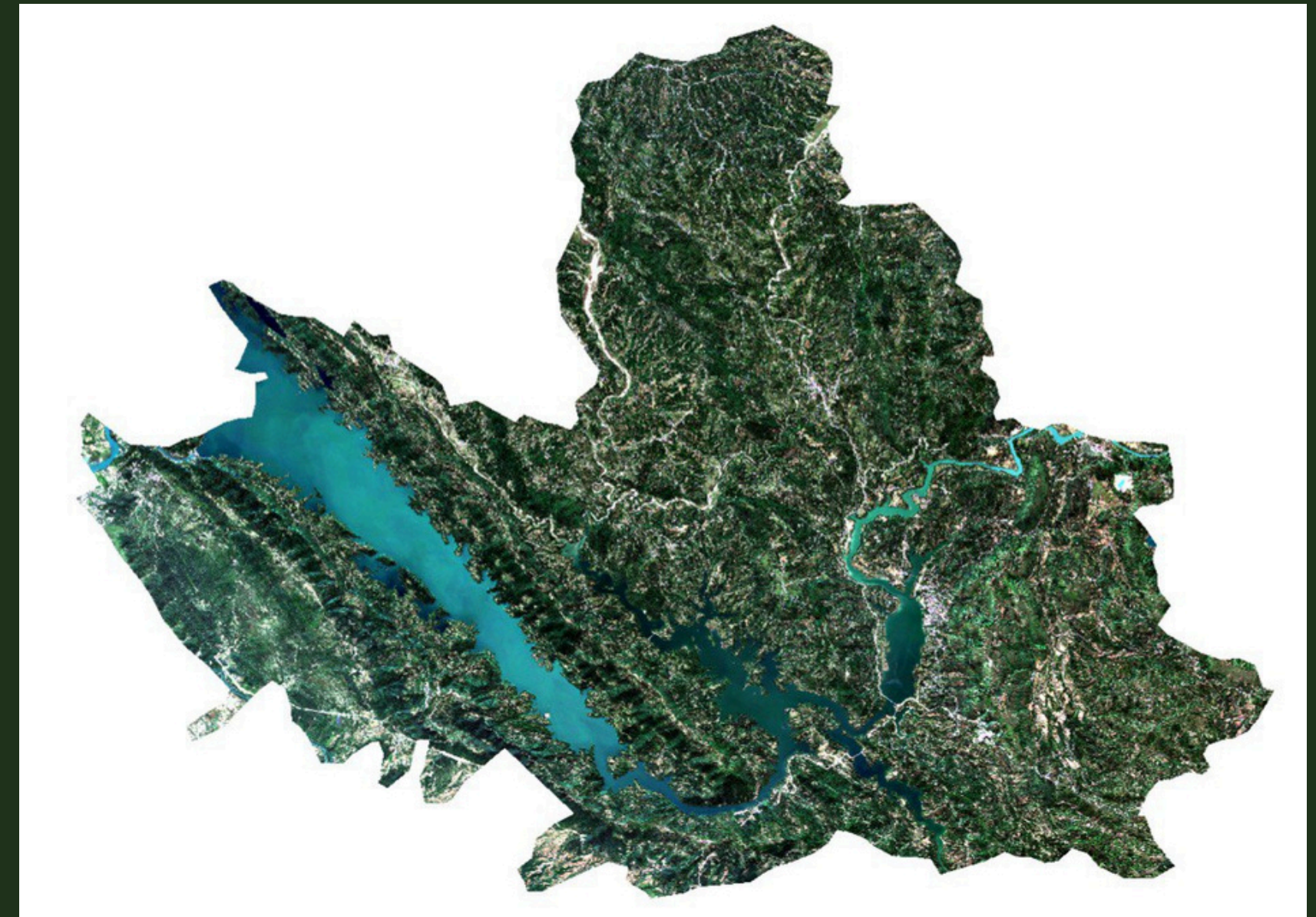
Datasets

Sentinel 2 Imagery

Source: Google Earth Engine



Pre Event

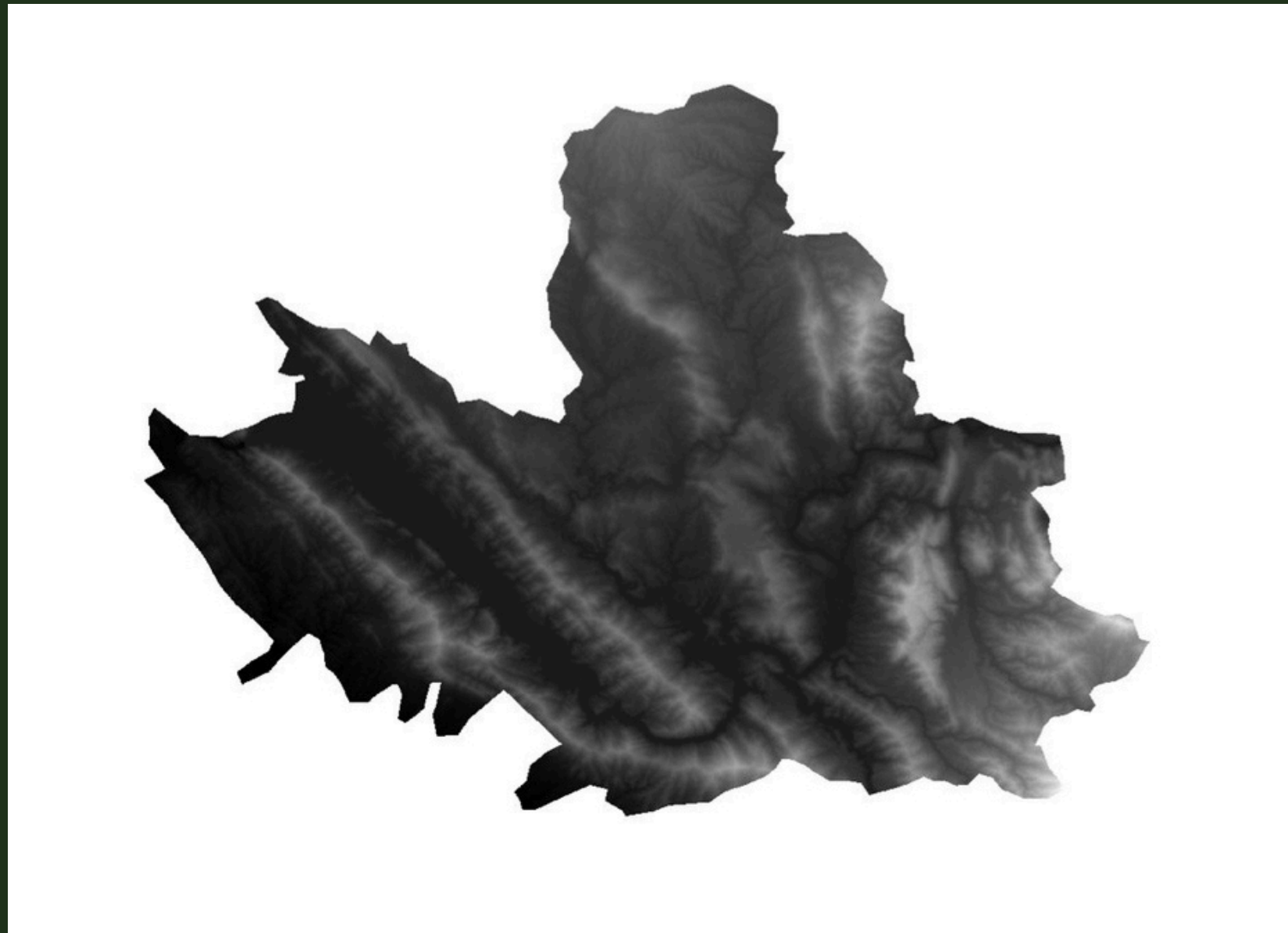


Post Event

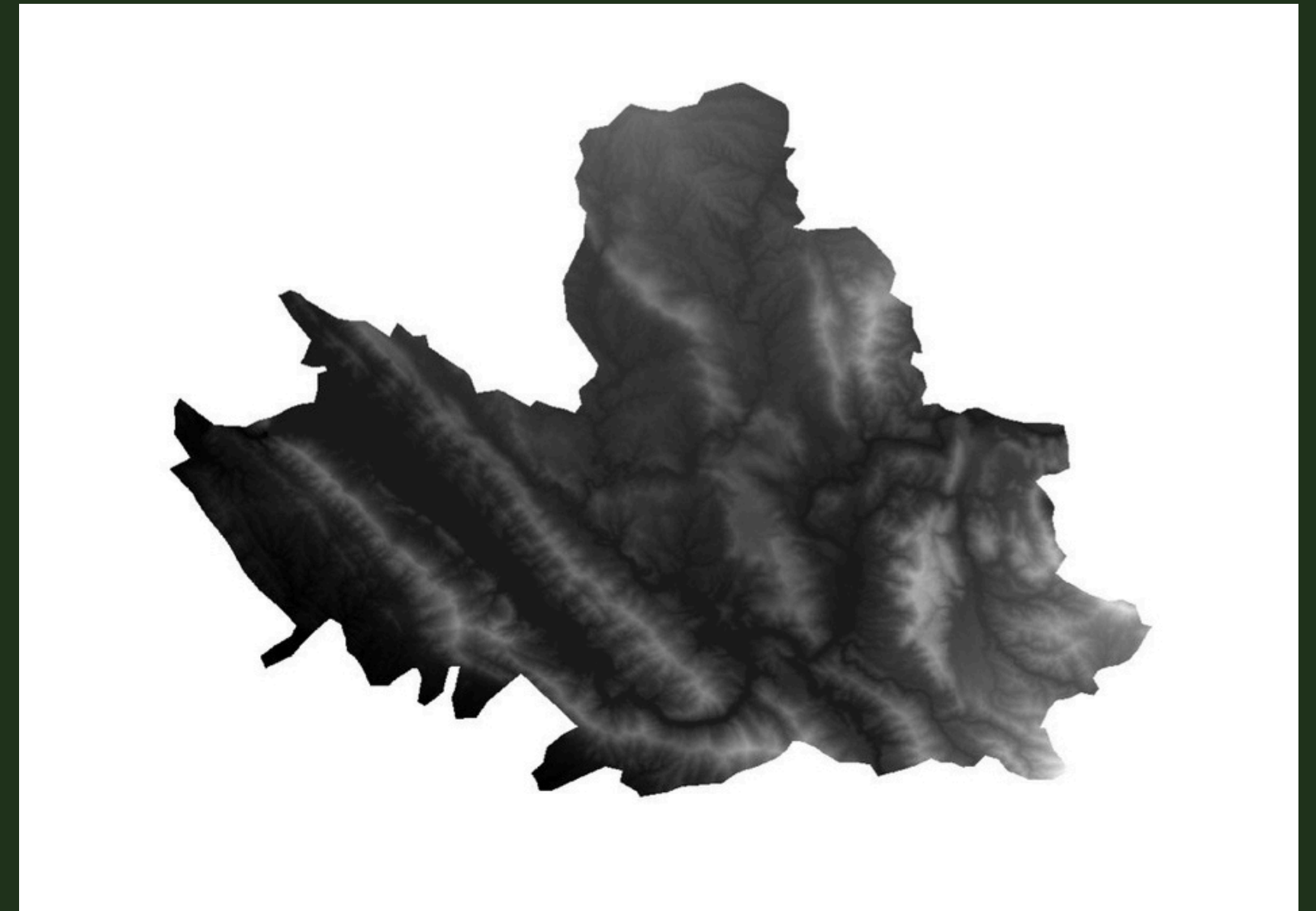
Datasets

Digital Elevation Model

Source: Google Earth Engine



Pre Event



Post Event

Indices

- **Normalized Difference Vegetation Index (NDVI)**

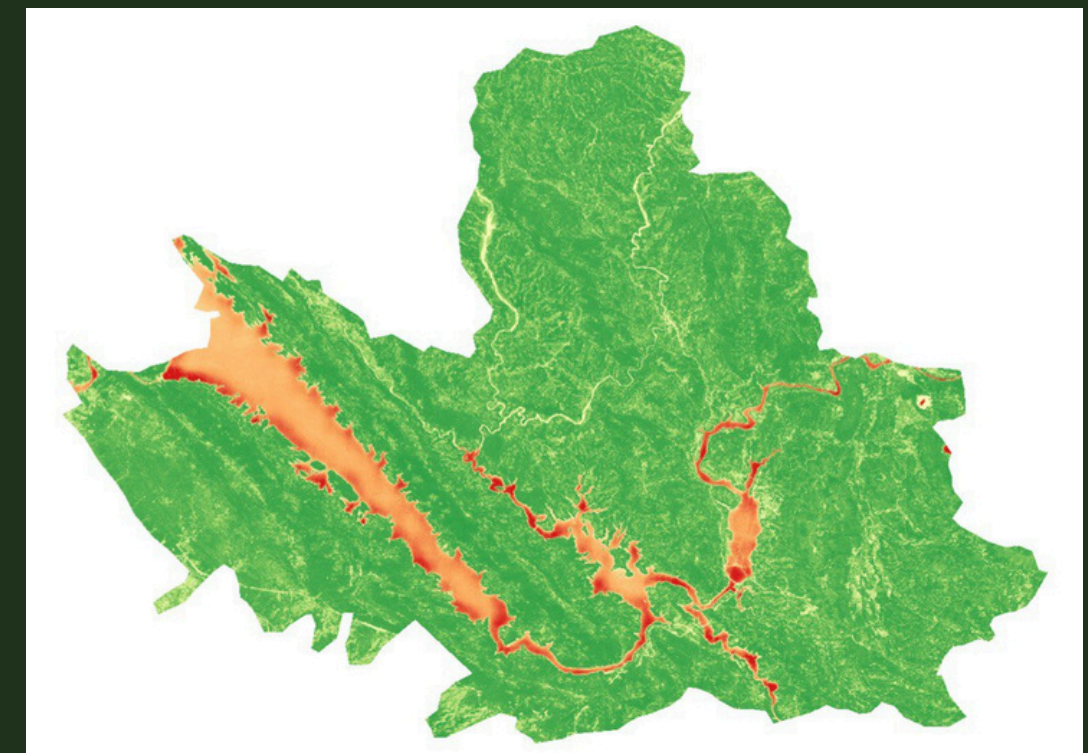
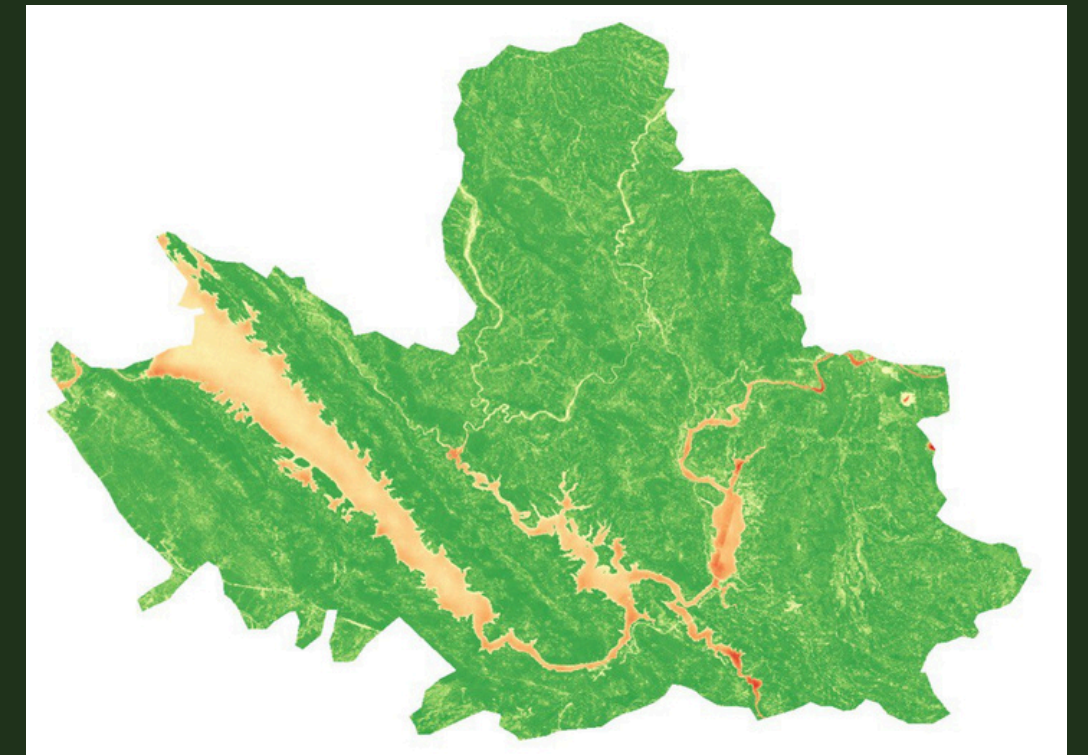
NDVI is a widely used remote-sensing index to assess vegetation health, density, and growth condition.

It is based on the principle that healthy vegetation strongly reflects near-infrared (NIR) radiation and absorbs red light for photosynthesis.

NDVI values range from -1 to +1, where higher positive values represent dense, healthy vegetation, and values near zero or negative indicate bare soil, water, or built-up areas.

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

Higher NDVI values indicate healthier, denser vegetation.



Indices

- **Normalized Difference Moisture Index (NDMI)**

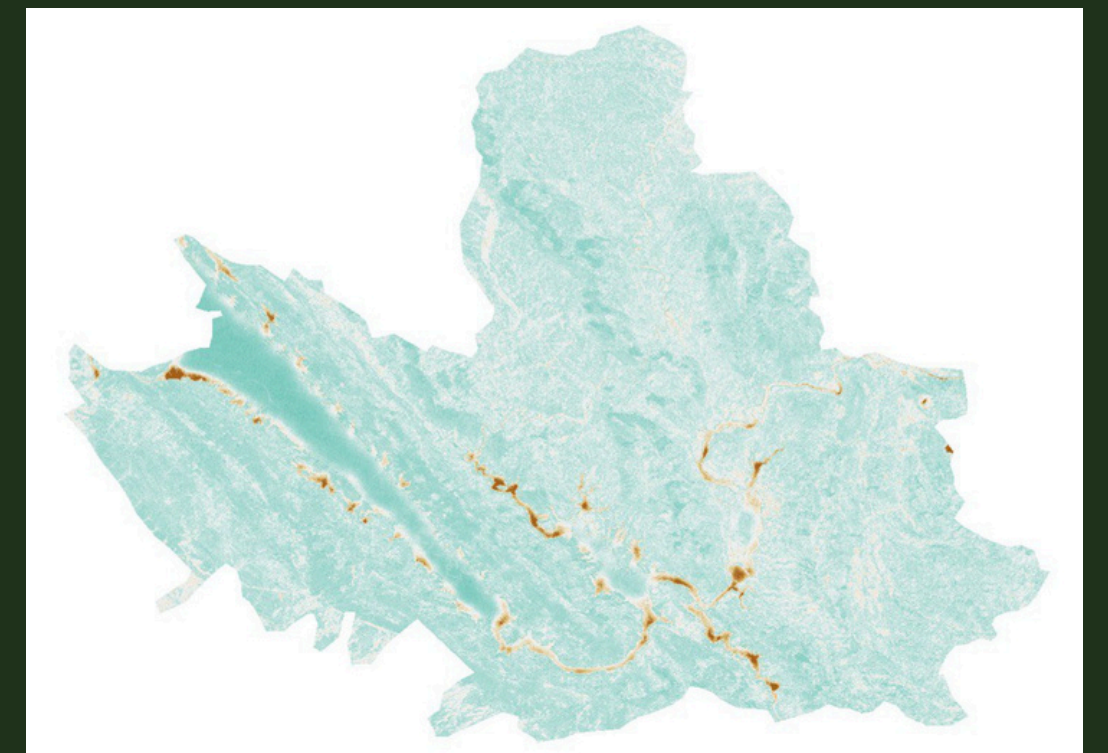
NDMI is used to assess vegetation and soil moisture content using satellite imagery.

It relies on the contrast between near-infrared (NIR), which vegetation reflects, and shortwave infrared (SWIR), which is sensitive to moisture.

NDMI values range from -1 to +1, where higher values indicate higher moisture levels and lower values indicate dry or stressed vegetation.

$$\text{NDMI} = (\text{NIR} - \text{SWIR}) / (\text{NIR} + \text{SWIR})$$

NDMI is widely used for drought assessment, crop water stress analysis, and vegetation moisture monitoring.



Indices

- **Normalized Difference Water Index (NDWI)**

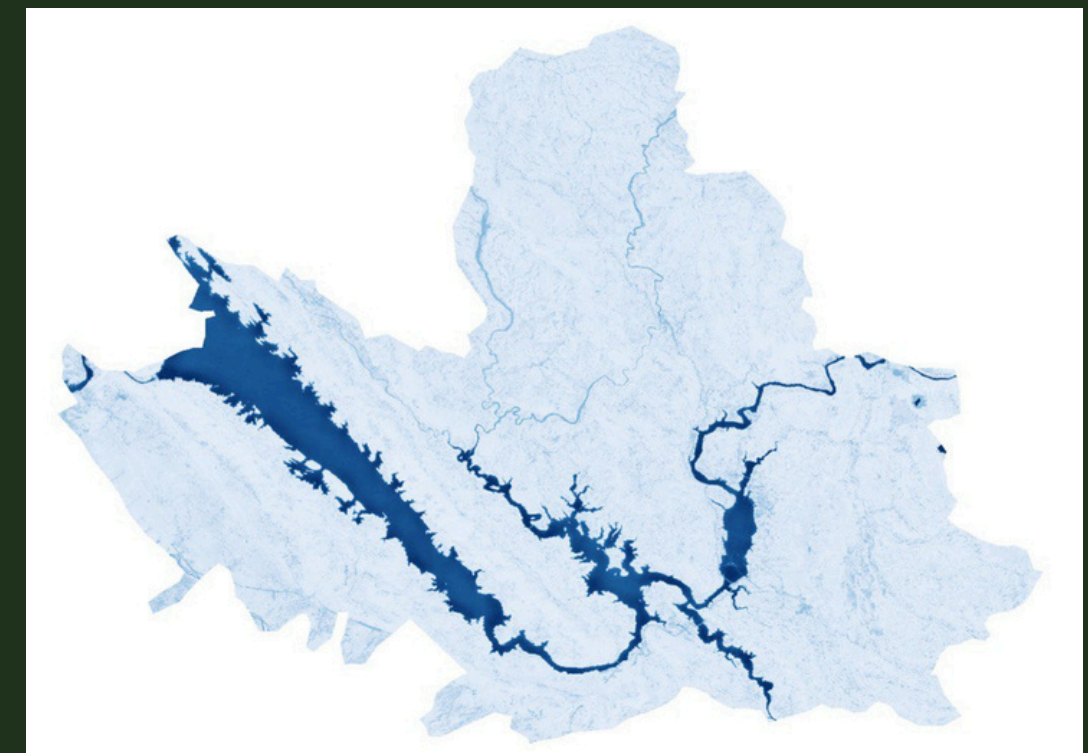
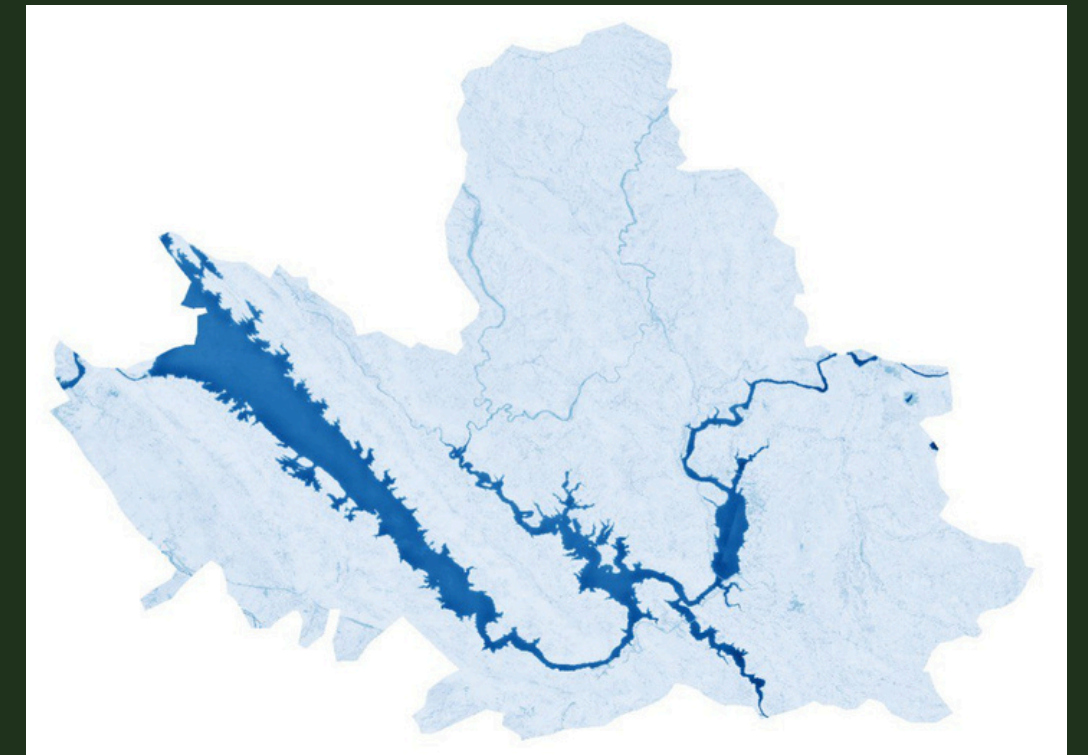
NDWI (Normalized Difference Water Index) is used to identify and monitor surface water and vegetation water content from satellite imagery.

It exploits the contrast between green light, which water reflects, and near-infrared (NIR) light, which water strongly absorbs.

NDWI values range from -1 to +1, where higher positive values indicate open water and lower values represent land, soil, or vegetation.

$$\text{NDWI} = (\text{Green} - \text{NIR}) / (\text{Green} + \text{NIR})$$

NDWI is commonly applied in water-body mapping, flood monitoring, and wetland analysis.



Indices

- **Bare Soil Index (BSI)**

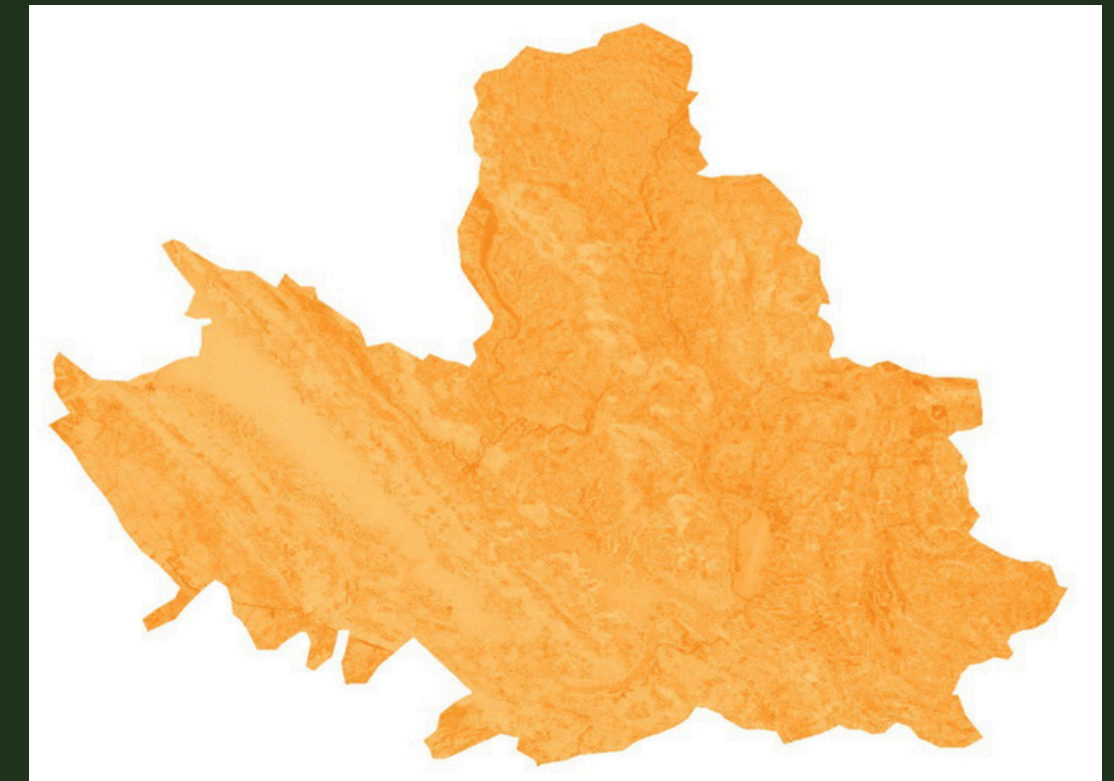
BSI is used to identify and map bare soil and exposed land surfaces using satellite imagery.

It combines blue, red, near-infrared (NIR), and shortwave infrared (SWIR) bands to enhance soil reflectance while suppressing vegetation and water signals.

Higher BSI values indicate bare or sparsely vegetated soil, while lower values indicate vegetation or water.

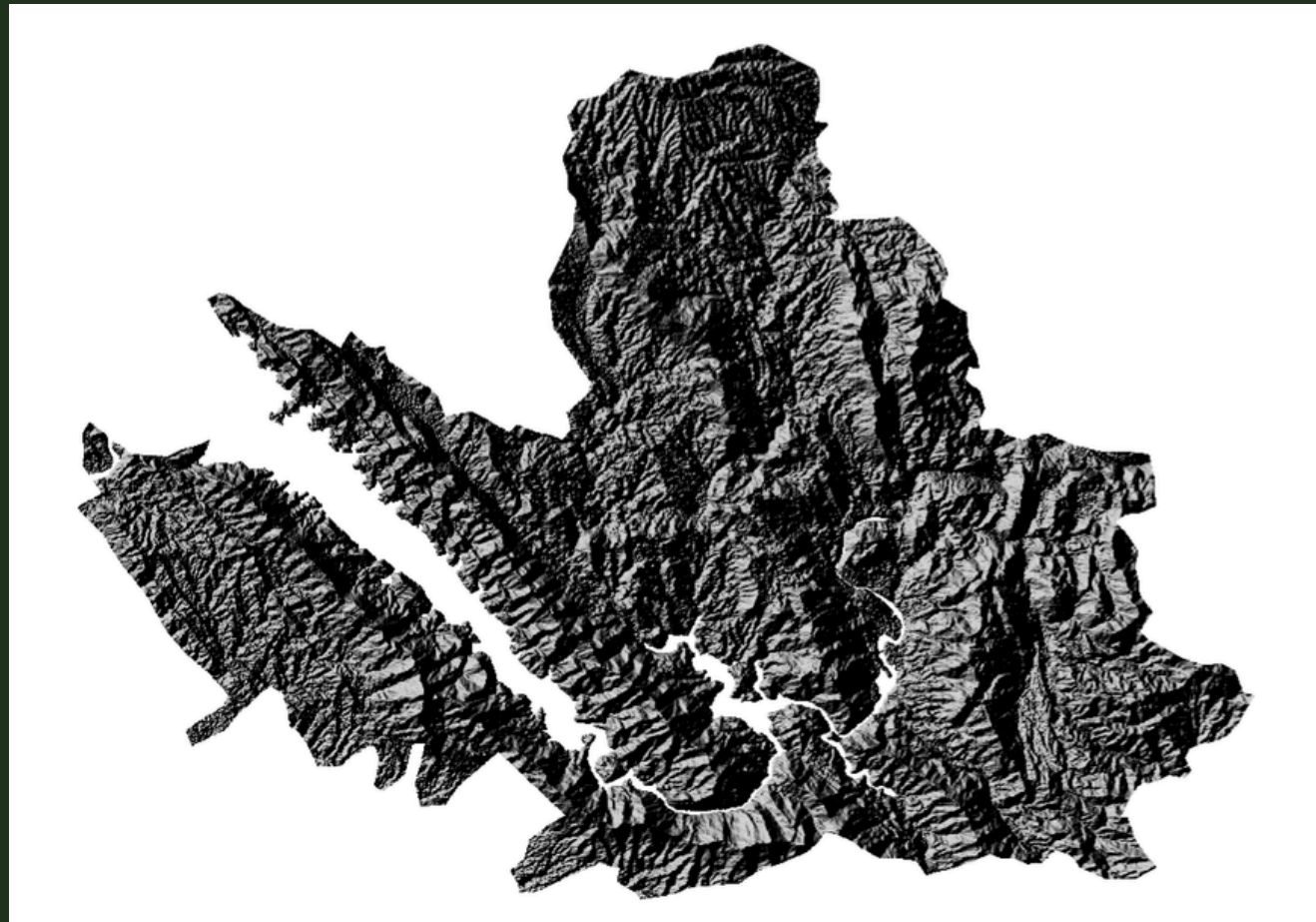
$$\text{BSI} = \frac{[(\text{SWIR} + \text{Red}) - (\text{NIR} + \text{Blue})]}{[(\text{SWIR} + \text{Red}) + (\text{NIR} + \text{Blue})]}$$

BSI is commonly used for soil exposure mapping, land degradation studies, and urban expansion analysis.

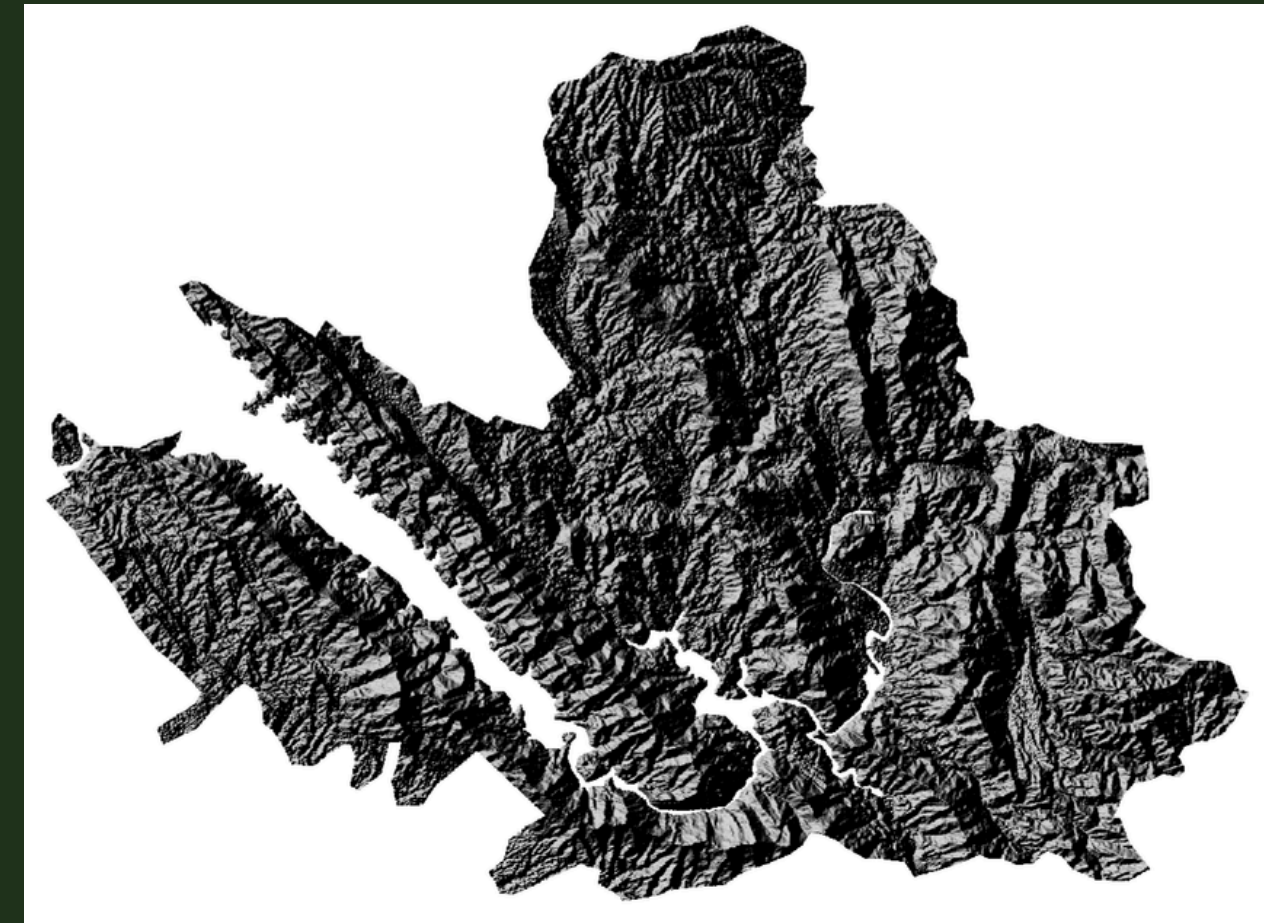


Topographic Analysis

Before

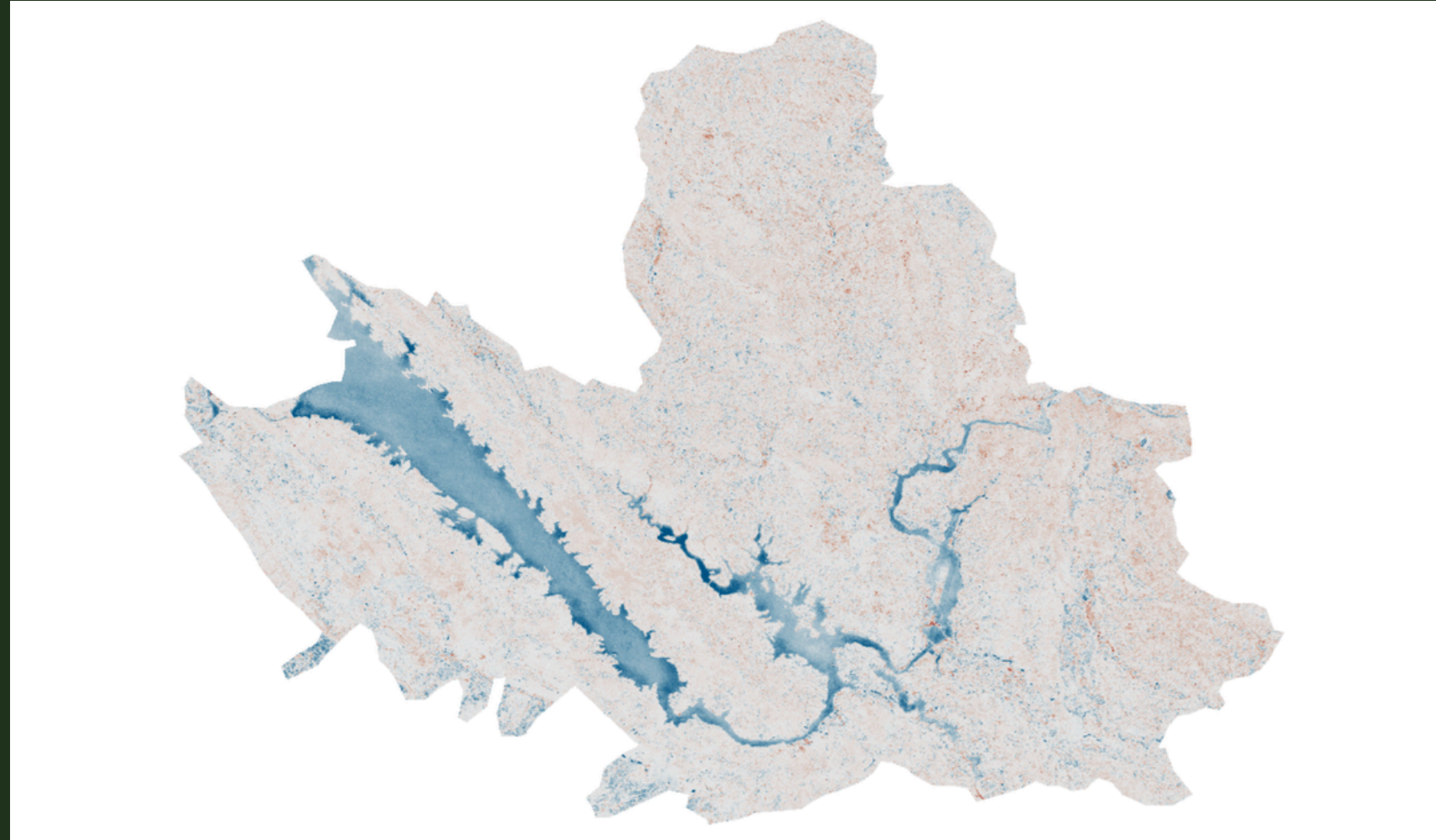


After

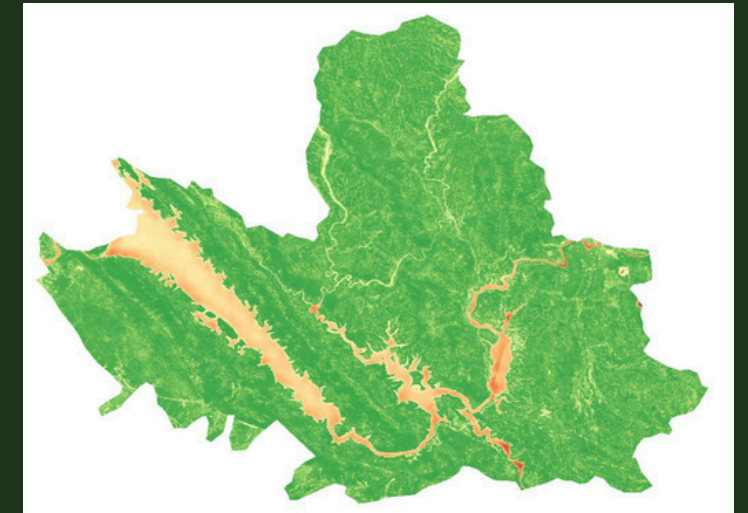


- Hillshade maps utilize the Digital Elevation Model (DEM) to simulate sunlight and shadows, providing a clear 3D visualization of the terrain's relief.
- Landslide susceptibility in the Bilaspur region is heavily governed by these specific steep slopes and rugged topographies.

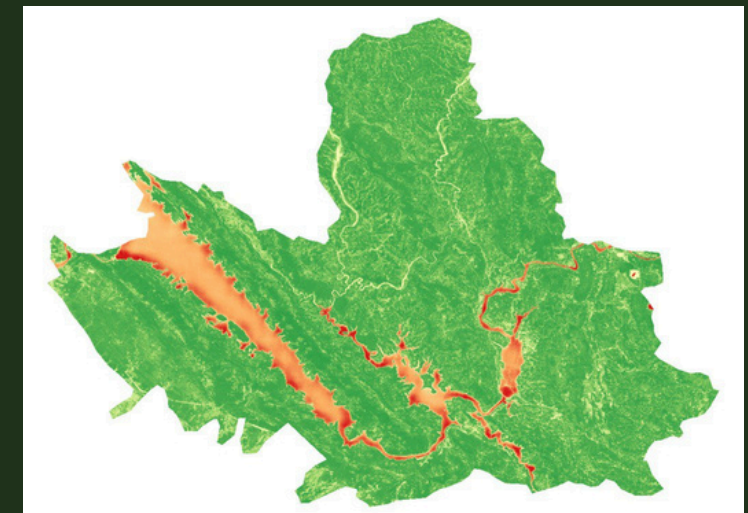
Assessing Vegetation Loss



Before



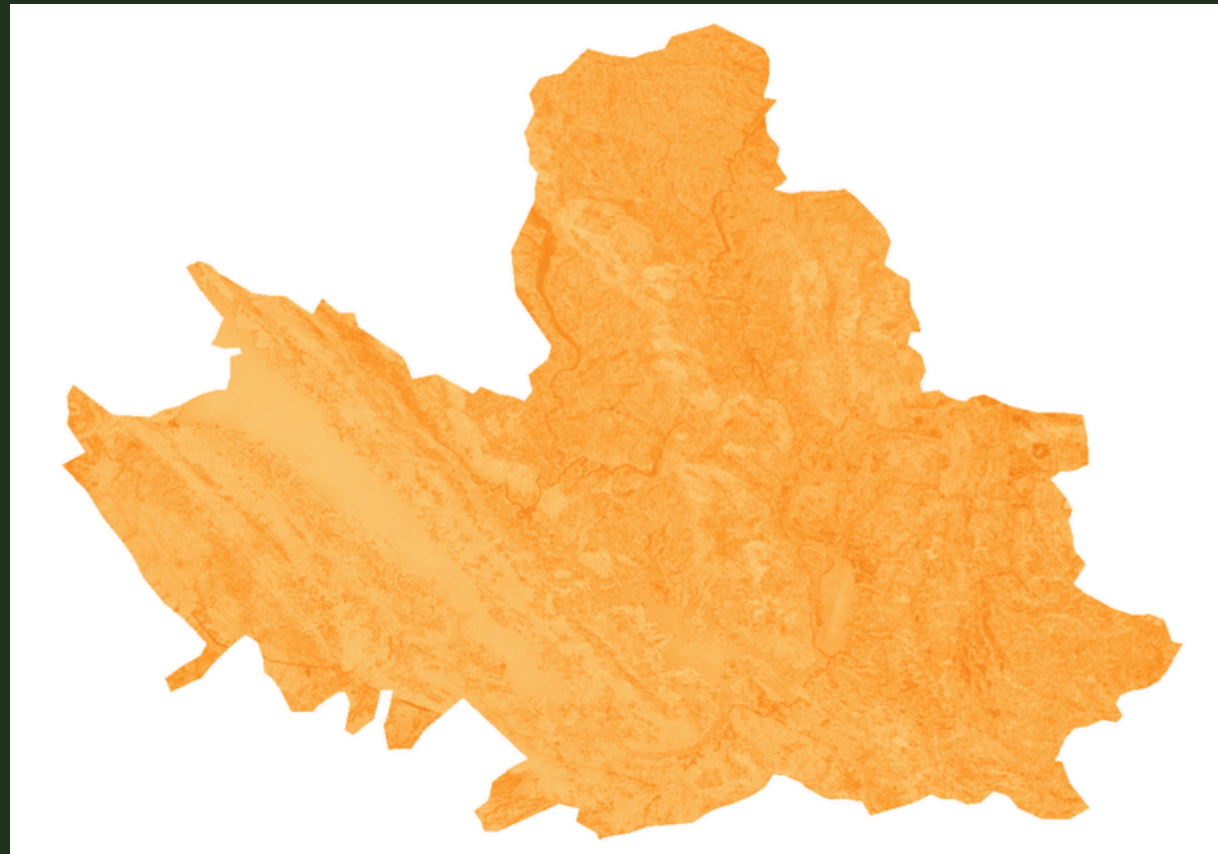
After



- Landslides completely strip away surface vegetation, creating a stark, sudden contrast in near-infrared reflectance.
- The sharp drop in NDVI values visibly outlines the boundaries of the landslide scar against the surrounding healthy forest

Assessing Soil Exposure

Before



After



- While vegetation drops during a landslide, bare earth exposure sharply spikes.
- The Bare Soil Index (BSI) highlights the newly exposed soil, underlying rock, and debris flow.
- The post-event BSI map shows a highly concentrated increase in values exactly where the slope failed, acting as a direct inverse confirmation to the NDVI drop.

Comparative Analysis

- **NDWI & NDMI:** While useful for identifying the background moisture conditions that trigger landslides, they struggle to map the actual post-event damage scar with sharp boundaries.
- **BSI:** Highly effective at identifying the newly exposed debris, but can sometimes falsely identify pre-existing barren land or roads as part of the landslide.
- **NDVI (Change Detection):** Emerges as the most accurate index. The sudden, localized drop in vegetation health provides the sharpest contrast, making it the superior metric for mapping the exact spatial extent of the affected area.

Conclusion

- Remote sensing and QGIS provide highly accurate, data-driven tools for assessing disaster zones like the October 7th Bhalughat event.
- By comparing multiple remote sensing indices, this study concludes that Change in NDVI, when verified alongside BSI, is the most reliable indicator for post-disaster mapping.
- Prioritizing NDVI change detection provides the highest accuracy in defining the damage perimeter, offering a critical and efficient tool for local authorities during post-disaster assessment and recovery planning



THANK YOU

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